# Bárbara Cruvinel Santiago, Ph.D.

*Email:* barbara.cruvinelsantiago@drexel.edu | barbara.santiago@aya.yale.edu *Website:* http://barbaracrusan.github.io/

**Education**

**Columbia University** – New York, NY, USA Aug 2018 – Jul 2023
Ph.D. in Physics (Jul 2023); M.Phil in Physics (May 2021); M.A. in Physics (May 2020)

**Yale University** – New Haven, CT, USA Aug 2013 – May 2017
B.S. in Physics (Intensive Track)

**Colégio Objetivo** (High School) – Santos, SP, Brazil Jan 2010 – Dec 2012

**Work and Research Experience**

**Visiting Assistant Professor of Physics at Drexel** – Philadelphia, PA Sep 2026 – Pres.
Teaching two undergraduate physics courses per quarter and pursuing nuclear security/astrophysics research.

**John B. Madden Dean of Berkeley College at Yale** – New Haven, CT Aug 2024 – Apr 2026
In-residence chief academic advisor to 450+ students, also serving on Yale College committees.

**Lecturer at the Yale Department of Physics** – New Haven, CT Aug 2024 – Apr 2026
Teaching undergraduate physics. Also affiliated faculty member of the *Yale Nuclear Security Program*.

**Post-Doctoral Researcher at Stanford (CISAC)** – Stanford, CA Sep 2023 – Jul 2024
Independent study examining dual-purpose technical research in order to aid in nuclear non-proliferation.

**Graduate Researcher at Columbia (Schiminovich Group)** – New York, NY Jan 2019 – Jul 2023
Work in Prof. Schiminovich group on the development of spectrographs for the study of intergalactic and circumgalactic media emission, and on the analysis of observational data.

**Post-Bacc Researcher at MIT-LIGO (Mavalvala Group)** – Cambridge, MA Jul 2017 – Jun 2018
Work in Prof. Nergis Mavalvala's LIGO group on a optomechanical light squeezing experiment. Focus on noise characterization and on the development of a light intensity stabilization servo.

**Undergraduate Researcher at Yale (DeMille Group)** – New Haven, CT Jun 2015 – May 2017
Work in Prof. David DeMille's group on the development and upgrade of hardware for his experiments.

**Undergraduate Researcher at CERN (Demers Group)** – Meyrin, Switzerland Jun 2014 – Aug 2014
Work in Prof. Sarah Demers' group at the ATLAS Experiment in the search for W,Z associated Higgs production.

**Most Important Honors and Awards**

**Stanton Nuclear Security Postdoctoral Fellowship** – USA 2023
Fellowship to pursue a post-doc at the Stanford Center for International Security and Cooperation (CISAC).

**Five Sigma Physicist Award** – USA 2021
American Physical Society Government Affairs award for outstanding advocacy.

**Physicists Coalition's Next-Generation Fellowship** – USA 2020
One of 4 fellows in the year-long program sponsored by the American Physical Society and the Princeton Science & Global Security Program to train early-career scientists in nuclear threat reduction policymaking.

**Future Investigators in NASA Earth and Space Science and Technology Fellowship** – USA 2019
NASA's 3-year graduate student fellowship awarded to student-designed research projects that contribute to NASA's exploration goals; formerly known as NESSF.

**Yale College Dean's Research Fellowship in the Sciences** – New Haven, USA 2016 and 2015
Summer award to do research at Yale and start senior thesis.

**Pierson College Richter Summer Fellowship** – New Haven, USA 2016 and 2015
Summer award to do research at Yale and start senior thesis.

**Tetelman Fellowship for International Research in the Sciences** – New Haven, USA 2014
Summer award to do research at CERN.

**Science, Technology and Research Scholars Program I (STARS I)** – New Haven, USA 2014
Selective mentoring program for minority students interested in STEM at Yale in their freshman year.

**Fundação Estudar Scholarship** – São Paulo, Brazil 2013
Merit-based scholarship awarded to 27 out of 11,000 applicants based on academic excellence and students' potential to lead and leave a positive legacy in Brazil.

**25th International Young Physicists' Tournament (IYPT)**, *Bronze Medal* – Germany 2012
IYPT is considered the Physics World Cup for high school students. Teams of 5 students selected in national competitions in each country work on open-ended problems and debate about them.

**24th International Young Physicists' Tournament (IYPT)**, *15th place* – Iran 2011
See description above.

**Skills**

*Programming:* **Python**, **Matlab**, **Mathematica**, **C/C++** and **HTML**.
*Languages:* **Portuguese** (native), **English** (fluent), **German** (intermediate) and **Spanish** (read).

**Leadership Positions**

**Yale STARS Advisory Board**, *Member* – New Haven, USA 2025 – Pres.
Residential college dean representative on the advisory board for the Yale Science, Technology, and Research Scholars (STARS) Program, which facilitates STEM undergraduate education.

**Yale College Admissions Committee**, *Member* – New Haven, USA 2025 – Pres.
Residential college dean representative serving on voting board 2-3 times per year.

**Yale Council of the Heads of College Finance Committee**, *Member* – New Haven, USA 2025 – Pres.
Residential college dean representative on task force to explore possible budget cuts due to endowment tax.

**Physicists Coalition for Nuclear Threat Reduction Fellowship Committee**, *Member* 2025 – Pres.
Committee that reviews Next-Generation Fellowship applications, project proposals and project outcomes. Members also serve as project advisors.

**Science Policy for Scientists Lecture Series**, *Co-Organizer* Spring 2023
Lecture series with experts in different policy areas targeting Columbia STEM grad students.

**SU(5) Peer Mentoring**, *Founding Member & Mentor/Facilitator* 2020-Pres.
SU(5) provides a network and a mentor to incoming astronomy and physics graduate students nationwide.

**Columbia Physics Grad Council**, *President & Diversity Liaison* – New York, USA 2019 – 2021
Served as President (2020-2021) and as Climate and Diversity Liaison (2019-2020). Addressed students' quality of life and academic issues alongside the dept.'s chair and committees, and other student groups.

**Columbia Spectra**, *Graduate Coordinator* – New York, USA 2018 – 2020
Formerly the Columbia Society for Women in Physics. I coordinated events targeting current and prospective grad students, and advised female and underrepresented undergrads in physics.

**Yale Physics Climate and Diversity Committee**, *Undergrad Rep.* – New Haven, USA 2015 – 2017
Represented the interest of Yale physics undergrads with respect to the climate of the department.

**Yale Women in Physics**, *Co-President & Co-Coordinator* – New Haven, USA 2014 – 2016
Served as President (2015-2016) and as Conference Co-Coordinator (2014-2015). Responsible for organizing a national conference, panels and social events for the Yale physics community.

**Brazil Club at Yale**, *President & Vice-President* – New Haven, USA 2014 – 2016
Served as President (2015-2016) and Vice-President (2014-2015). Responsible for organizing social events, talks and meals for people interested in Brazil at Yale.

**Yale Admissions Office**, *Science Tour Guide* – New Haven, USA Summer 2015
Worked part-time as a tour guide of Yale's science facilities.

**Extra Coursework and Summer Schools**

**Princeton School on Science and Global Security** – Princeton, NJ, USA Oct 2022
Selective 4-days school on nuclear security issues for early-career scientists.

**George Washington University Nuclear Security Boot Camp** – Online Jun 2020
Selective week-long course on nuclear security taught by science/policy experts and policymakers.

**Dunlap Institute Introduction to Astronomical Instrumentation** – Toronto, Canada Aug 2016
Selective week-long theoretical/experimental summer school at the Dunlap Institute for Astronomy and Astrophysics at the University of Toronto for college seniors and graduate students.

**Course of General Astronomy** – Brazil Nov 2011 – May 2012
Course for the top 120 in the Brazilian Astronomy Olympiad competing to represent Brazil internationally.

**IV Advanced School of Nuclear Energy (EAEN)** – São Paulo, Brazil Jul 2011
Selective week-long school for high schoolers at the Institute for Nuclear Energy Research at USP.

**Teaching, Mentoring and Outreach**

**PHYS 101/102: Fundamentals of Physics I/II**, *Professor* – Drexel, Philadelphia, USA Fall 2026
Large-format intro physics lecture courses with associated labs required for engineering majors.

**PHYS 3200: Physics and Public Policy**, *Lecturer* – Yale, New Haven, USA Fall 2025
My own course on the connection between technical topics in physics and policymaking. More on my website.

**Introductory Physics Recitations and Labs**, *TA* – Columbia, New York, USA 2018-2020
Responsible for teaching introductory physics recitations and labs to over 140 pre-medical and engineering undergrads over 2 years. Courses: UN1202, UN1401, UN1292, UN1291.

**National Selection Tournament of US IYPT**, *Juror* – Philadelphia, USA Spring 2018
Judged selection of high school team to represent the US at the International Young Physicists' Tournament.

**New Haven Science Fair**, *Mentor* – New Haven, USA Spring 2017

Mentored a New Haven public school 9th grader on a thermodynamics project. She won 2nd place in the entire fair and an award for science communication.

**Yale College Dean's Office**, *STARS Peer Mentor* – New Haven, USA 2016 – 2017
Worked as Peer Mentor for incoming minority students interested in pursuing STEM majors at Yale who participate in the Science, Technology and Research Scholars Program I.

**Yale Girls Science Investigations**, *Volunteer* – New Haven, USA Fall 2016
Volunteered to help GSI, which brings elementary school girls to Yale to do fun experiments for a day.

**Yale Office of International Students and Scholars**, *Peer Liaison* – New Haven, USA 2015 – 2016
Worked as a Peer Liaison to help incoming international freshmen adjusting to the Yale environment.

**Yale Center for Language Studies**, *Portuguese Tutor* – New Haven, USA Mar 2015 – Apr 2017
Worked as a tutor for Yale students studying Portuguese language.

**Splash at Yale**, *Teacher* – New Haven, USA Spring 2015
Co-taught the course *The Physics of Interstellar* for high school students.

**Brazilian Student Council on Undergraduate Education**, *Mentor* – Brazil 2014 – 2015
Mentored 3 Brazilian high school students applying to American colleges.

**Yale Department of Spanish and Portuguese**, *Portuguese Grader* – New Haven, USA 2013 – 2014
Worked as a grader of homework assigned in Portuguese classes.

## Scientific Publications

**(In preparation)** B. Cruvinel Santiago, N. Melso, I Cevallos Aleman, B Smiley, H. R. Ong, D. Schiminovich. *Kinematics of the Fireworks Galaxy: Processing and Analysis of NGC 6946 Images Obtained with the Circumgalactic Hydrogen-Alpha Spectrograph.* Submitting in 2026.

**(In preparation)** B. Cruvinel Santiago, N. Melso, I Cevallos Aleman, B. Smiley, H. R. Ong, D. Schiminovich. *The Little Dumbbell Case: Using the Circumgalactic Hydrogen-Alpha Spectrograph to Analyze Kinematics in Nebulae.* Submitting in 2026.

N. Melso, D. Schiminovich, M. Sitaram, I. Cevallos Aleman, B. Cruvinel Santiago, B. Smiley, H. R. Ong,. *Very Extended Ionized Gas Discovered around NGC 1068 with the Circumgalactic Hα Spectrograph*. The Astrophysical Journal (2024).

N. Melso, D. Schiminovich, B. Smiley, H. R. Ong, B. Cruvinel Santiago, M. Sitaram, I. Cevallos Aleman, S. Graber, M. Murillo, M. Rosenthal, and I. Stelea. *The Circumgalactic Hα Spectrograph (CHαS) I. Design, Engineering, and Early Commissioning*. The Astrophysical Journal (2022).

## Conferences and Presentations

**Yale Kimball Smith Series/Back from the Brink** – New Haven, CT, USA, Oct. 2025, *Invited Panelist.* Panel: *The Urgent Threat of Nuclear Weapons: How the Yale Community Can Make a Difference*.

**Yale Kimball Smith Series/Yale Nuclear, Particle and Astrophysics Joint Seminar** – Yale Department of Physics/Wright Lab, New Haven, CT, USA, Sep. 2024, *Oral Presentation.* B. Cruvinel Santiago, *Flagging Dual-Purpose Research in the Physical Sciences*.

**CISAC Seminar Series** – Stanford Center for International Security and Cooperation, Stanford, CA, USA, May 2024, *Oral Presentation.* B. Cruvinel Santiago, *Flagging Dual-Purpose Research in the Physical Sciences*.

**International Studies Association Annual Convention 2024** – San Francisco, CA, USA, Apr. 2024, Session on Nuclear Weapons and Emerging Technologies, *Oral Presentation.* B. Cruvinel Santiago, *Identifying dual-purpose academic research in the physical sciences for the benefit of the nuclear non-proliferation regime.*

**The Increasing Danger of Nuclear Weapons: How Physicists Can Help Reduce the Threat** – International Center for Theoretical Physics (ICTP), Trieste, Italy, Oct. 2023, *Oral Presentation.* B. Cruvinel Santiago, *Advocacy in Brazil in the Nuclear Field (Past and Present).*

**Stanton Nuclear Security Seminar** – Stanton Foundation, Washington, D.C., USA, Oct. 2023, *Oral Presentation.* B. Cruvinel Santiago, *Flagging Dual-Purpose Research in the Physical Sciences.*

**IEEE Ethics** – IEEE, Online, May 2023, *Invited Panelist.* Panel: *The Ethics of Weapons Technology Development*.

**Princeton School on Science and Global Security**, Princeton, NJ, USA, Oct. 2022, *Oral Presentation.* B. Cruvinel Santiago, *Brazil Shouldn't Let Bolsonaro Get in the Way of the Ban Treaty.*

**Astronomy on Tap**, New York, NY, USA, Sep. 2022, *Oral Presentation.* B. Cruvinel Santiago, *Missing: have you seen these baryons? And how can we find them?*.

**EstroGênias: Meninas na Ciência** – Tecnologia Educacional (Grupo Positivo), Online, Mar. 2021, *Invited Panelist.* Panel: *How Schools Should Promote STEM Education*.

**I Congresso Nacional de Inovação e Popularização da Ciência** – Federal University of Minas Gerais (UFMG), Online, Oct. 2020, *Invited Panelist.* Panel: *Science, Innovation and Market after COVID-19*.
**235$^{th}$ American Astronomical Society Meeting** – Honolulu, HI, USA, Jan. 2020, *Poster Presentation.* B. Cruvinel Santiago, N. Melso, H. R. Ong, B. Smiley, M. Rosenthal, M. Murillo, S. Graber, and D. Schiminovich, *Circumgalactic H-alpha Spectrograph: Initial Results from the First Engineering Runs*.
**MIT Physics Exploration Day** – Cambridge, MA, USA, Apr. 2018, *Poster Presentation.* B. Cruvinel Santiago, and I. Phinney, *Overview of Advanced LIGO and future methods of noise reduction*.
**Scientista Symposium** – New York, NY, USA, Apr. 2017, *Poster Presentation.* B. Cruvinel Santiago, and D. DeMille, *Evaluating and Upgrading a Fast Compact Laser Shutter Prototype*.
**Pierson College Mellon Forum at Yale** – New Haven, CT, USA, Feb. 2017, *Oral Presentation.* B. Cruvinel Santiago, *Development of Hardware for Atomic Physics Experiments.*
**APS Conference for Undergraduate Women in Physics (CUWiP) at Yale** – New Haven, CT, USA, Jan. 2015, *Conference Co-Coordinator.*

## Congressional Action

**APS-led Congressional Action**, *Physicists Coalition Delegation Member* Summer 2021
Met with congressional offices to advocate for "no first use" of nuclear weapons.
**NAGPS Legislative Action Days (LAD)**, *Columbia Delegation Member* Mar 2020
Met with congressional staff about issues that affect grad students (online because of COVID-19).
**Lobbying against ICE removal of SEVP exemption**, *Initiative Organizer* Jul 2020
After ICE removed the exemption permitting international students to maintain their status taking online courses, I led a group of physics students who met (virtually) with House Reps. and congressional staff.

## Other Honors and Awards

**Michele Dufault Memorial Scholarship for Yale Women in STEM** – New Haven, USA 2016
Academic year scholarship awarded to selected women pursuing STEM majors at Yale.
**Physics Olympiad of the State of São Paulo 2012 (OPF)**, *Bronze Medal* – Brazil 2012
OPF is a state-level physics competition for São Paulo high school students.
**3rd Brazilian Linguistics Olympiad (OBL)**, *Honorable Mention* – Brazil 2012
OBL is a national linguistics competition for Brazilian high school students.
**Rio Leadership Institute (ILRIO) Prep Program**, Mentee – Brazil 2012
Mentoring program that selects 30 out of ∼1000 Brazilian high school students based on academic merit.
**IYPT Brasil 2012**, *Silver Medal, Selected for national delegation* – Brazil 2012
National competition that selects 5 high school students to go to represent Brazil at the IYPT.
**XV Brazilian Olympiad on Astronomy and Astronautics (OBA)**, *Gold Medal* – Brazil 2012
OBA is a national astronomy competition for Brazilian middle and high school students.
**Regional Tournament of Exact Sciences (TEX)**, *1st place* – Santos, Brazil 2012
TEX is a competition on physics, chemistry and math for students in the Baixada Santista region.
**IYPT Brasil 2011**, *Silver Medal, Selected for national delegation* – Brazil 2011
National competition that selects 5 high school students to go to represent Brazil at the IYPT.
**XIV Brazilian Olympiad on Astronomy and Astronautics (OBA)**, *Silver Medal* – Brazil 2011
OBA is a national astronomy competition for Brazilian middle and high school students.
**Physics Olympiad of the State of São Paulo 2012 (OPF)**, *Honorable Mention* – Brazil 2010
OPF is a state-level physics competition for São Paulo high school students.
**IYPT Brasil 2010**, *Bronze Medal* – Brazil 2010
National competition that selects 5 high school students to go to represent Brazil at the IYPT.
**XII Brazilian Olympiad on Astronomy and Astronautics (OBA)**, *Silver Medal* – Brazil 2009
OBA is a national astronomy competition for Brazilian middle and high school students.
**XI Brazilian Olympiad on Astronomy and Astronautics (OBA)**, *Silver Medal* – Brazil 2008
OBA is a national astronomy competition for Brazilian middle and high school students.
**Objetivo Scholarship Exam**, *Full-Ride Scholarship* – Brazil 2008
Merit-based scholarship for 5 years of school at Colégio Objetivo.

## Op-Eds and Articles

B. Cruvinel Santiago, L. Rand *Beyond the cinematic feat: Consequences Oppenheimer ignored*, Physics Today, July 2024.
B. Cruvinel Santiago, *Why Brazil shouldn't let Bolsonaro stand in the way of ratifying the nuclear ban treaty*, Bulletin of the Atomic Scientists, April 2022.
B. Cruvinel Santiago, *A female Ph.D. student's cautionary tale and the need for peer mentorship*, Women

in Astronomy, Summer 2020.
B. Cruvinel Santiago, *Undergraduate Profile Katie Melbourne (SY '19): From Yale to NASA*, Yale Scientific Magazine, Spring 2017.
B. Cruvinel Santiago, *Caro fantasma do preconceito inconsciente... (Dear ghost of implicit bias...)*, Accent (The Undergraduate Multilingual Magazine at Yale), Fall 2016.
B. Cruvinel Santiago, *Em trânsito: navegando entre dois mundos (In Transit: Navigating Between Two Worlds)*, Accent (The Undergraduate Multilingual Magazine at Yale), Spring 2016.
B. Cruvinel Santiago, *Agressividade infundada: o novo vício do eleitor brasileiro (Non-sense aggression: the new addiction of the Brazilian voter)*, Accent (The Undergraduate Multilingual Magazine at Yale), Fall 2014.
B. Cruvinel Santiago, *O Brasil e o império da ciência (Brazil and the science empire)*, Accent (The Undergraduate Multilingual Magazine at Yale), Spring 2014.